

LD11 Source Coverage Checklist

Detailed proof that MVD and uploaded-source topics are covered individually.

Field	Value
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Topic	Learning Day 11 - Hyperparameter Tuning
Document type	Individual PDF documentation
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Submission note
This is an individual PDF document. The documentation package intentionally does not include a combined/merged PDF.

1. Coverage Matrix

Topic	Location	Coverage type	Evidence	Status
Grid Search	Main + extended	Practical	Bank RF, KNN, Logistic, Ridge	Covered
Random Search	Main notebook	Practical	Bank RF, XGBoost	Covered
Nested CV	Main notebook	Theory + practical	Nested CV explanation and run	Covered
Bayesian optimisation	Main + PDF theory	Practical + theory	BayesSearchCV plus deeper acquisition theory	Covered
KerasTuner RandomSearch	Main notebook	Practical	Santander NN tuning	Covered
Hyperband	Main notebook	Practical	Santander NN Hyperband	Covered
KNN tuning	Extended notebook	Practical	GridSearchCV with scaling	Covered
Logistic Regression tuning	Extended notebook	Practical	GridSearchCV C/penalty	Covered
Ridge Regression tuning	Extended notebook	Practical	alpha search on regression dataset	Covered
CASH	Extended notebook	Practical + theory	Model family comparison	Covered
Hyperopt	Extended notebook	Practical safe workflow	Objective/search space/trials	Covered
Optuna	Extended notebook	Practical safe workflow	Study/trials/objective	Covered
Successive Halving	Extended notebook	Practical safe workflow	Multi-fidelity RF search	Covered
AutoML systems	Master theory PDF	Detailed theory	Auto-sklearn, TPOT, Auto-WEKA context	Covered
NAS deeper theory	Master theory PDF	Detailed theory	Search space, strategy, performance estimation	Covered
Multi-fidelity beyond Hyperband	Master theory PDF	Detailed theory	Low-resource early stopping logic	Covered

2. Coverage Interpretation

The Day 11 work is now divided correctly: the main notebook covers the MVD and neural-network tuning requirements, while the extended notebook and individual PDFs cover the missed uploaded-source topics. This prevents the MVD from being treated as the final boundary of learning.

Topics such as AutoML systems, NAS details, Bayesian acquisition functions, and multi-fidelity optimisation are not all required as full production implementations, but they are explained in detail because they appear in the broader source material and are important for conceptual understanding.

Coverage statement

The package covers both practical notebook work and supporting theory, with clear separation between implemented sections and theory-focused